

Leukapheresis

Leukapheresis is a process to remove extra white blood cells from the blood. It may be used if someone has very high levels of white blood cells when they are first diagnosed with leukaemia (leukemia).

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What is leukapheresis?

Some people have a very high number of white blood cells in their blood when they are diagnosed with [leukaemia](#) (also spelt leukemia). These cells can gather in the blood vessels and cause symptoms such as:

- breathlessness
- headaches
- blurred vision

Doctors can remove some white blood cells from the blood using a machine called a cell separator. This is called leukapheresis.

It may also be used for if you are pregnant when you are diagnosed.

Having leukapheresis

During leukapheresis, you lie on a bed or reclining chair. A small plastic tube ([cannula](#)) is put in a vein in each arm. Each cannula is connected to the cell separator by a tube. Blood goes from the arm through the tube into the cell separator.

As the blood travels through the cell separator, the machine removes the white blood cells. The rest of the blood and blood cells are then returned to the body through the cannula in the other arm. This takes a few hours.

Leukapheresis is painless, but some people may find it uncomfortable having the cannula put in.

Your haematology team can give you more information about this.

Glossary

Useful words to know

When you are dealing with leukaemia, you may come across lots of new words and not know what they mean.

Some of these words are explained here. If you need more information or support, you can call the Macmillan Support Line free on **0808 808 00 00**.

Blood cells

Cells which are found mainly in the blood. The 3 main types of blood cell are red blood cells, white blood cells and platelets.

Blood vessels

A network of tubes called arteries, veins and capillaries that carry blood around the body.

Cannula

A short, thin tube a doctor or nurse puts into a vein in the arm or the back of the hand. It can be used to give you fluids and treatment or to take blood.

Leukaemia

Cancer of the blood cells.

White blood cells

Cells in your blood that fight and prevent infection. There are several types of white blood cell. The 2 most important types are neutrophils and lymphocytes.

About our information

References



Below is a sample of the sources used in our chronic myeloid leukaemia (CML). If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

Smith G, Apperley J, Milojkovic D, et al. A British Society for Haematology Guideline on the diagnosis and management of chronic myeloid leukaemia. British Journal of Haematology, 2020; 191, 2, Available from <https://b-s-h.org.uk/> [accessed on October 2020].

Hochhaus A, Saussele S, Rosti G, et al. Chronic Myeloid Leukaemia: ESMO Clinical Practice Guidelines. Annals of Oncology, 2017; 28 (suppl 4), iv41-iv51. Available from <https://www.esmo.org/> [accessed on October 2020].

Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Senior Medical Editor, [Dr Anne Parker](#), Consultant Haematologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Date reviewed

Reviewed: 01 March 2022 | Next review: 01 March 2025

ⓘ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

[Get help](#)